Lines, angles and Polygons

Worksheet 2

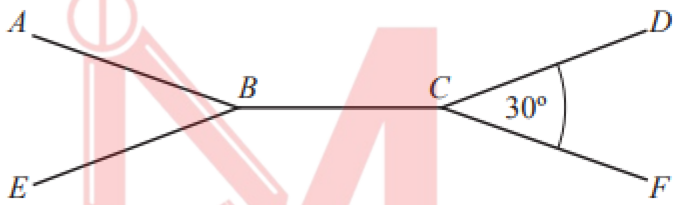
1. W16/qp11/q7

Each interior angle of a regular polygon is 171°. Find the number of sides of the polygon. [2]

2. S17/qp12/q19

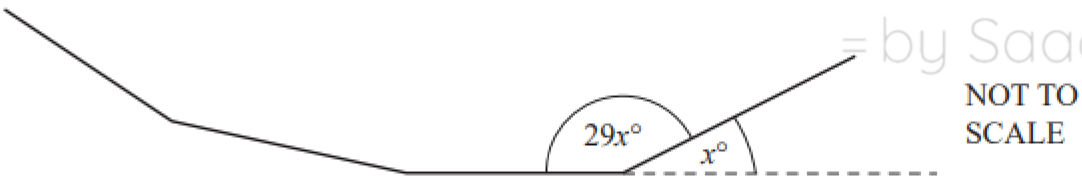
a. A pentagon has four angles of $2x$° and one angle of x°. Calculate the value of x. [2]

b. ABCD and EBCF are parts of two identical regular n-sided polygons.



The two polygons are joined together along edge BC. Angle $D\hat{C}F$ = 30°. Calculate the value of n. [2]

3. W17/qp22/q17 (0580)



The diagram shows part of a regular polygon. The exterior angle is x°. The interior angle is $29x$°. Work out the number of sides of this polygon. [3]

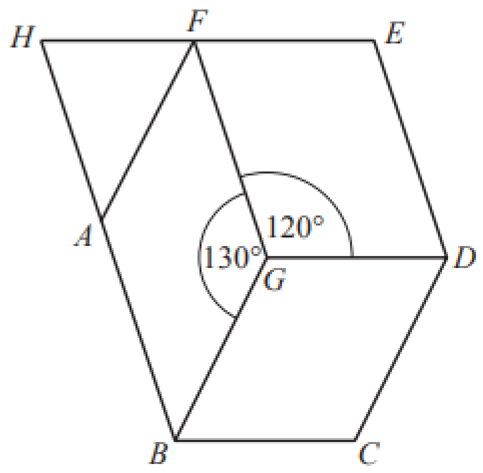
4. S18/qp12/q14

An irregular polygon has 22 sides.

a. Calculate the sum of all its interior angles. [2]

b. Two of the angles in the polygon are each 170°. The remaining 20 angles are equal to each other. Calculate the size of one of the 20 equal angles. [2

5. W18/qp11/q9



In the diagram, $AFGB$, $BGDC$ and $FEDG$ are parallelograms. EF and BA produced meet at H. $B\hat{G}F$ = 130° and $D\hat{G}F$ = 120°.

a. Find $B\hat{G}D$. [1]

b. Find $A\hat{B}G$. [1]

c. Find $H\hat{F}G$. [1]

d. Find $F\hat{H}A$. [1]

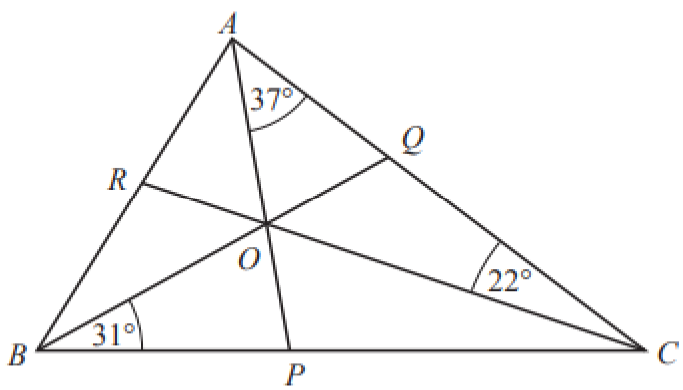
6. W18/qp11/q18

Four interior angles of a hexagon are 100°, 110°, 120° and 140°.

The remaining two interior angles are equal to each other.

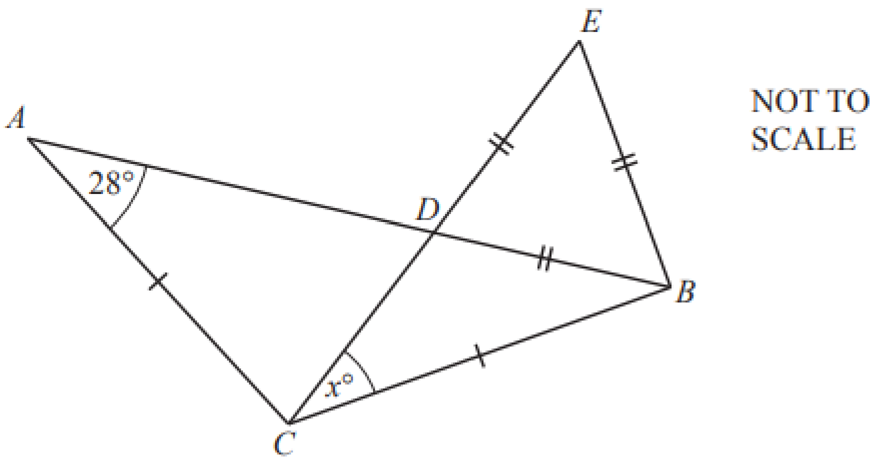
Calculate the size of one of these interior angles. [3]

7. W19/qp12/q16(a)



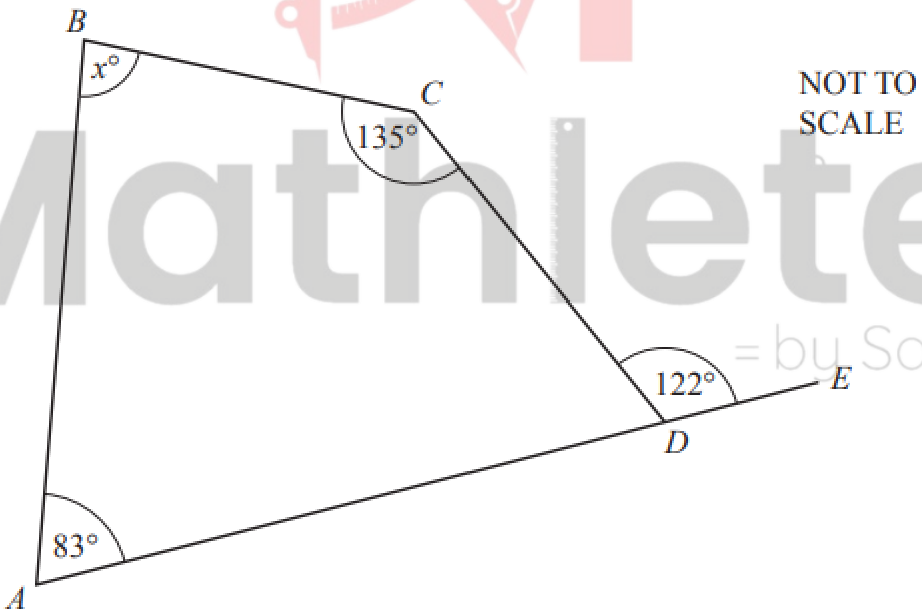
In the diagram, AP, BQ and CR are the bisectors of the angles of triangle ABC. The bisectors intersect at O. $O\hat{B}P$ = 31° , $O\hat{C}Q$ = 22° and $O\hat{A}Q$ = 37° . Calculate $P\hat{O}C$. [1]

8. S20/qp11/q5



The diagram shows an isosceles triangle ABC and an equilateral triangle BDE. D is the intersection of AB and CE . Angle $B\hat{A}C$ = 28°. Calculate x. [2]

9. S21/qp11/q5



The diagram shows quadrilateral $ABCD$ with AD extended to E. Angle $B\hat{C}D$ = 135°, angle $B\hat{A}D$ = 83° and angle $C\hat{D}E$ = 122°. Find the value of x. [2]

Answer Key:

1. 40
2. (a)60 (b)24
3. 60
4. (a)3600 (b)163
5. (a)110° (b)50° (c)120° (d)60°
6. 125
7. 59°
8. 32
9. 84